Network switching subsystem - GSM flowchart

SYSTEM OPERATIONS / GSM CORE NETWORK

Network Switching Subsystem

Classic GSM • Circuit-switched registration, call routing and mobility

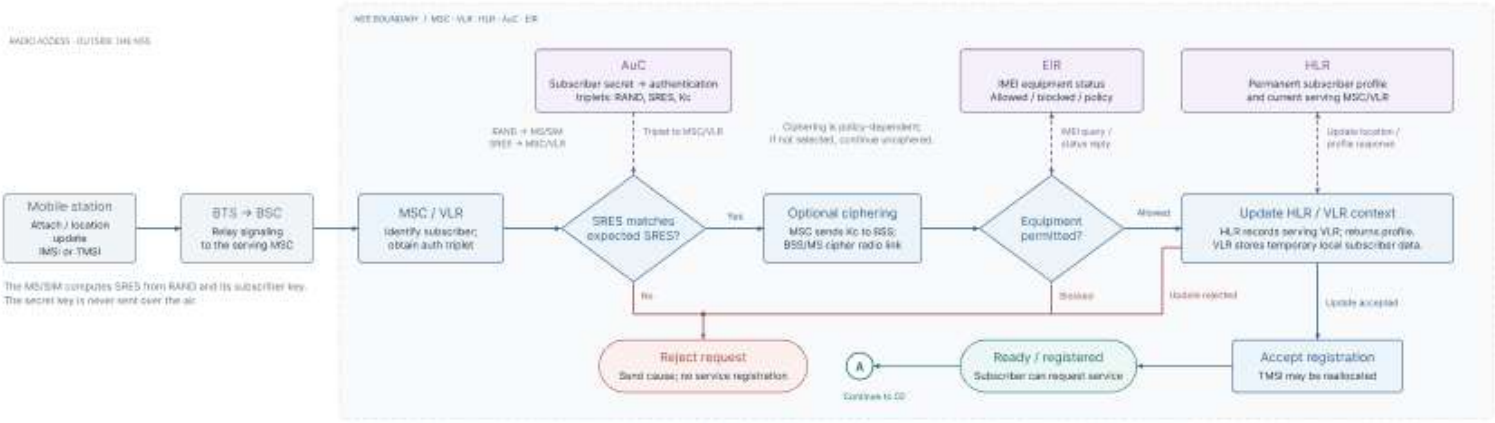
Read 01 → 02. Follow labeled arrows; matching circles link the registration and call flows.

HOW TO READ

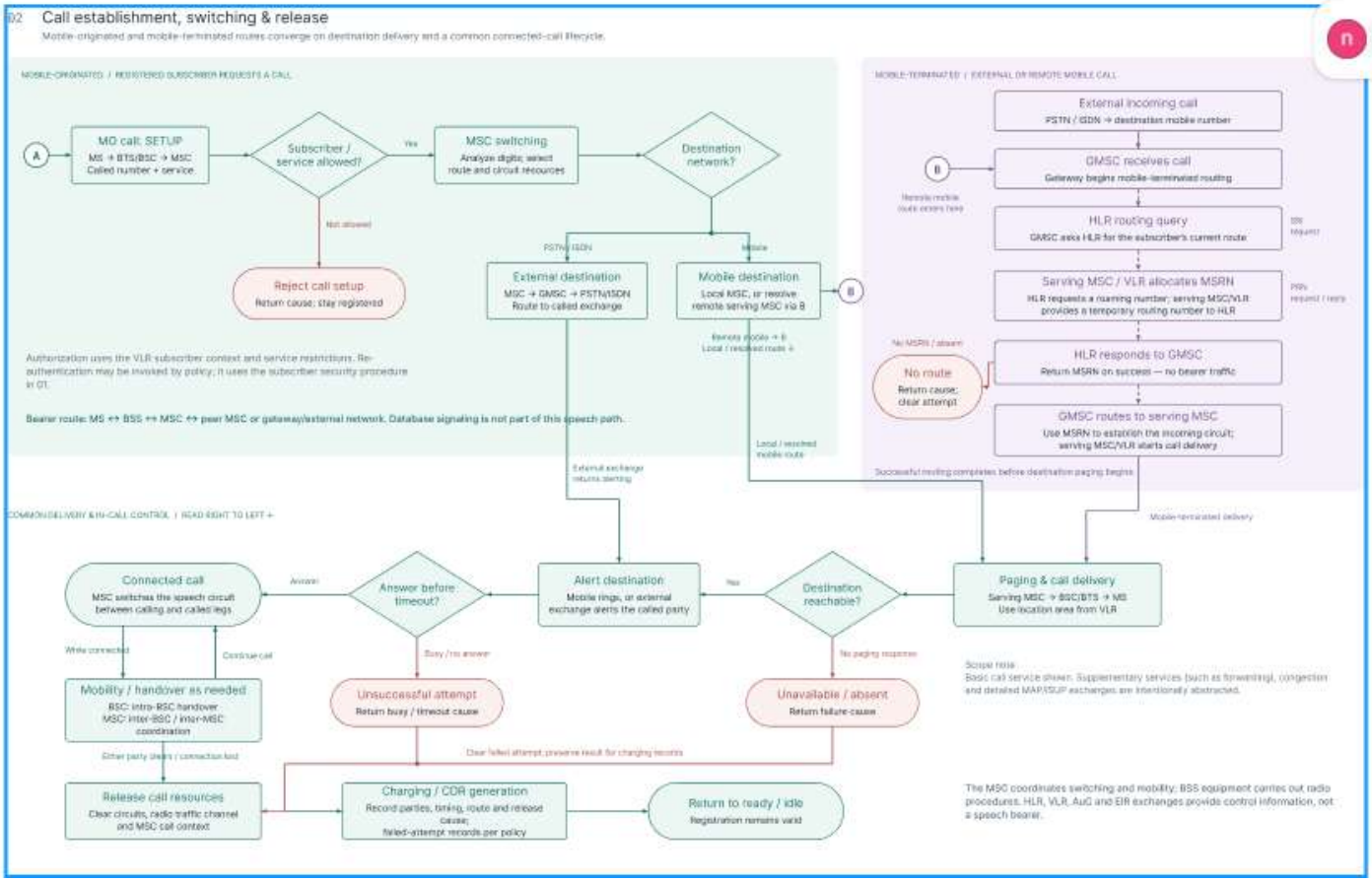
- Operational sequence / call-control flow
- Signaling / database request or response
- Non-success / release path

01 Access, security & registration

Location update / attach establishes a valid subscriber context before circuit-switched service.



Security roles stay separate: AuC validates subscriber credentials; EIR checks the handset; HLR/VLR manage subscriber and location data. None carries bearer traffic.


02 Call establishment, switching & release
Mobile-originated and mobile-terminated routes converge on destination delivery and a common connected-call lifecycle.
MOBILE-ORIGINATED / REGISTERED SUBSCRIBER REQUESTS A CALL
A
MO call: SETUP
MS → BTS/BSC → MSC
Called number + service
Subscriber / service allowed?
Yes
Not allowed
MSC switching
Analyze digits; select route and circuit resources
Destination network?
PSTN / ISDN
Mobile
Reject call setup
Return cause; stay registered
External destination
MSC → GMSC → PSTN/ISDN
Route to called exchange
Mobile destination
Local MSC, or receive remote serving MSC via B
B
Authorization uses the VLR subscriber context and service restrictions. Re-authentication may be invoked by policy; it uses the subscriber security procedure in 01.
Bearer route: MS ↔ BSS ↔ MSC ↔ peer MSC or gateway/external network. Database signaling is not part of this speech path.
MOBILE-TERMINATED / EXTERNAL OR REMOTE MOBILE CALL
External incoming call
PSTN / GSN → destination mobile number
B
GMSC receives call
Gateway begins mobile-terminated routing
HLR routing query
GMSC asks HLR for the subscriber's current route
SRI request
Serving MSC / VLR allocates MSRN
HLR requests a roaming number; serving MSC/VLR provides a temporary routing number to HLR
PRN request / reply
No MSRN / absent
No route
Return cause; clear attempt
HLR responds to GMSC
Return MSRN on success — no bearer traffic
GMSC routes to serving MSC
Use MSRN to establish the incoming circuit; serving MSC/VLR starts call delivery
Successful routing completes before destination paging begins
Mobile-terminated delivery
External exchange returns alerting
Local / resolved mobile route
COMMON DELIVERY & IN-CALL CONTROL / READ RIGHT TO LEFT →
Paging & call delivery
Serving MSC → BSC/BTS → MS
Use location area from VLR
Destination reachable?
No paging response
Alert destination
Mobile rings, or external exchange alerts the called party
Answer before timeout?
Answer
Busy / no answer
Connected call
MSC switches the speech circuit between calling and called legs
While connected
Continue call
Mobility / handover as needed
BSC: intra-BSC handover
MSC: inter-BSC / inter-MSC coordination
Either party clears / connection lost
Unsuccessful attempt
Return busy / timeout cause
Unavailable / absent
Return failure cause
Clear failed attempt; preserve result for charging records
Release call resources
Clear circuits, radio traffic channel and MSC call context
Charging / CDR generation
Record parties, timing, route and release cause; failed-attempt records per policy
Return to ready / idle
Registration remains valid
Scope note
Basic call service shown. Supplementary services (such as forwarding), congestion and detailed MAP/SUP exchanges are intentionally abstracted.
The MSC coordinates switching and mobility; BSS equipment carries out radio procedures. HLR, VLR, AuC and EIR exchanges provide control information, not a speech bearer.